

Unsupervised Neural Machine Translation for Low-Resource Domains via Meta-Learning

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19 Feb, 2023

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Conclusions

- Unsupervised machine translation
 - * Achieved comparable performance against supervised machine translation
 - suffers from data-scarce domains
 - by utilizing only a small amount of training data
 - extend the meta learning algorithm
 - * To address a low-resource challenge for UMT
 - ✓ utilize the meta learning approach

Proposed Approach

01

- Problem Setup
 - n out-domain datasets ($D_{out} = \{D^{0out}, \dots, D^{nout}\}$)
 - D_{in} indicates an in-domain dataset (not included in D_{out}) : a target domain
 - both D_{out} and D_{in} is assumed to be composed of unpaired language corpora
 - finetune the UNMT model (*by minimizing both the losses Language modeling and Back-translation with D_{in} .)

Proposed Approach

01

- MetaUMT
 - uses two training phases (the meta-train and the meta test)
 - During the meta-train phase
 - the model first learns domain-specific knowledge (i.e., adapted parameters)
 - obtain ϕ_i for each i -th out-domain dataset by using one-step gradient descent
 - In the meta-test phase
 - the model learns the adaptation by optimizing θ with respect to ϕ_i .
 - to update θ using each ϕ_i learned from the meta-train phase•

Proposed Approach

01

- MetaGUMT

- cause the model to overfit (since a small amount of training data)
- not utilizing high-resource domain knowledge

➤ **Objective:** incorporate high-resource domain knowledge
and generalizable knowledge into the model parameters

✓ meta-train loss

✓ cross-domain loss

Algorithm A.1 MetaGUMT

Require: α, β : step sizes

- 1: Pretrain θ by using XLM
 - 2: **while** not done **do**
 - 3: **for** all $\mathcal{D}_{out}^i$ **do**
 - 4: Evaluate $\nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^{lm}(\theta)$ with respect to source and target language sentences from $\mathcal{D}_{out}^i$
 - 5: **Back-translation** generates source and target language sentences using the current translation model
 - 6: Evaluate $\nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^{bt}(\theta)$ with using pseudo-generated sentences
 - 7: Sum each gradient:
 $\nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^s = \nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^{lm}(\theta) + \nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^{bt}(\theta)$
 - 8: Compute adapted parameters with one-step gradient descent:
 $\phi^i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^s(\theta)$
 - 9: **end for**
 - 10: Update $\theta \leftarrow \theta - \beta \nabla_{\theta}(\mathcal{L}_{cd} + \mathcal{L}_{ag})$
 - 11: **end while**
-

Proposed Approach

01

- MetaUMT vs MetaGUMT
 - Meta-train phase

$$\phi^i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{D}_{out}^i}^s(\theta)$$

$$\mathcal{L}_{\mathcal{D}_{out}^i}^s = \mathcal{L}_{\mathcal{D}_{out}^i}^{lm}(\theta) + \mathcal{L}_{\mathcal{D}_{out}^i}^{bt}(\theta)$$

MetaGUMT : this phase is exactly same with the meta-train phase of MetaUMT

- Meta-test phase

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{i=0}^n \mathcal{L}_{\mathcal{D}_{out'}^i}^s(\phi^i)$$

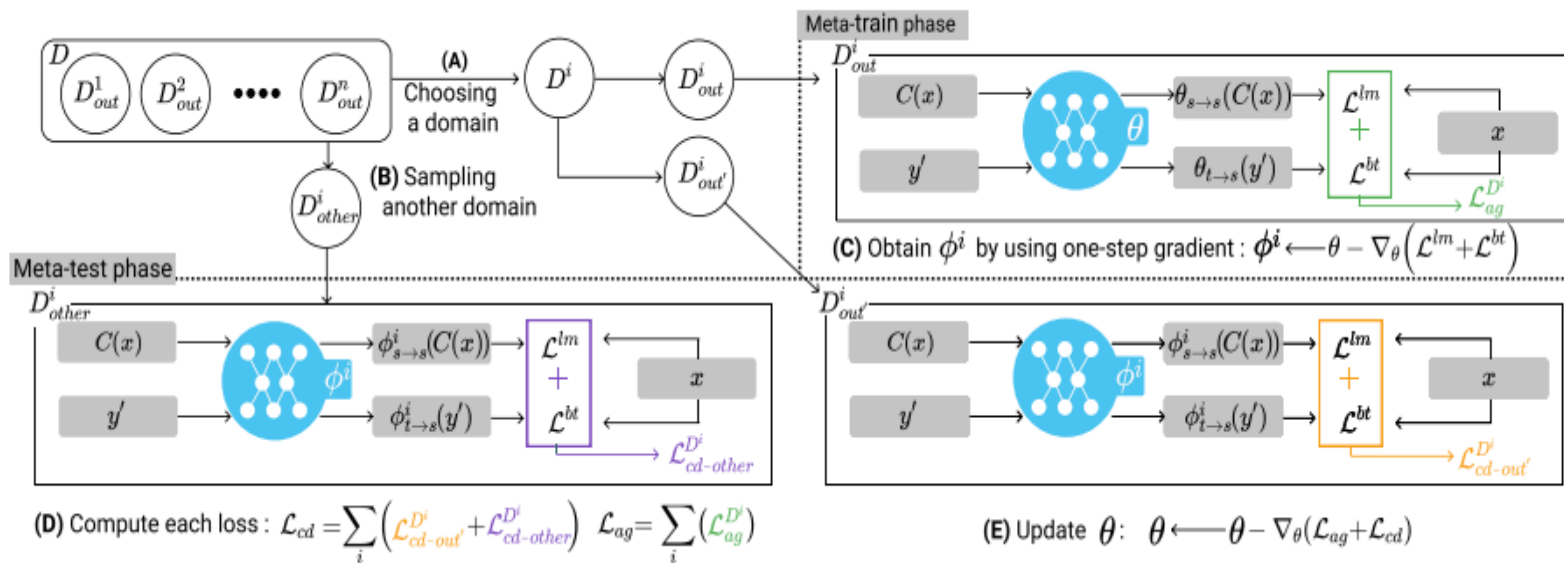
$$\theta \leftarrow \theta - \beta \nabla_{\theta} (\mathcal{L}_{cd} + \mathcal{L}_{ag}).$$

the sum of the two of our losses

Proposed Approach

01

- Training process of our proposed MetaGUMT



- UNMT
 - **Unsupervised neural machine translation** (ref. <https://arxiv.org/abs/1710.11041>)
 - Instead of not using parallel corpus, a significant number of monolingual sentences (1M-3M sentences)
 - the prerequisite limits : low-resource domains
- Meta learning
 - **Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks**(ref. <https://arxiv.org/abs/1703.03400>)
 - for handling a small amount of training data
 - the previous studies : on a supervised model that requires labeled corpora

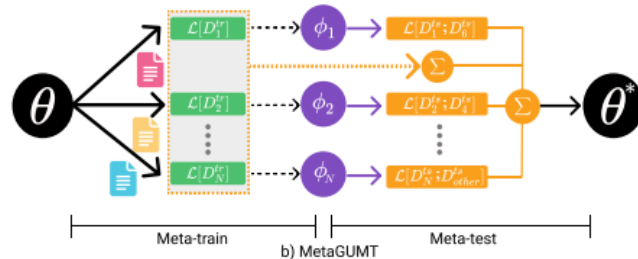
- MetaUMT
 - a new meta-learning approach for UNMT
 - to find the optimal initialization for the model parameters that can adapt to a new domain
 - even with only a small amount of monolingual data
- 1. the meta-train phase : adapts model parameters to a domain
- 2. the meta-test phase: optimizes the parameters obtained from the meta-train phase
- 3. After obtaining optimally initialized parameters : fine-tune the model using a target domain (i.e., a low-resource domain)

- MetaGUMT

- * finding optimally initialized parameters

that incorporate high-resource domain knowledge and generalizable knowledge

1. discards meta-train knowledge used to update adapted parameters in the meta-train phase
2. instead of validating the same domain used in the meta-train phase
3. inject generalizable knowledge into the initial parameters by utilizing another domain in the meta-test phase.



- Low-Resource Neural Machine Translation (1)
 - the performance of NMT models depends on the size of the parallel dataset
 - To address this problem, utilizing monolingual datasets.
 - apply dual learning, back-translation
 - pretraining the model with bilingual corpora
 - the UNMT methods without using any parallel corpora
 - incorporating methods such as BPE and cross-lingual representations (following those of the supervised NMT)
 - * require plenty of monolingual datasets

- Low-Resource Neural Machine Translation (2)
 - Transferring the knowledge from high-resource domains to a low-resource domain
 - applicable in specific scenarios
 - * To address the issues
 - ✓ define a new task as the unsupervised domain adaptation on the low resource dataset

- Meta Learning
 - Given a small amount of training data
 - prone to overfitting
 - failing to find a generalizable solution
 - To find the optimal initialization of the model parameters to a low-resource dataset
 - To address the low-resource UNMT by exploiting meta-learning approaches

- Unsupervised Neural Machine Translation
 - Initialization : XLM (cross-lingual language model)
 - Language modeling : use a denoising auto encoder
 - Back-translation : the model learn the mapping functions

θ : parameterized the NMT model

x and y : source and the target sentences (from S and T)

S and T : source and a target monolingual language dataset

Experiments

03

- Dataset and Preprocessing
 - Experiments on eight different domains
 - OPUS 4 (Tiedemann, 2012)

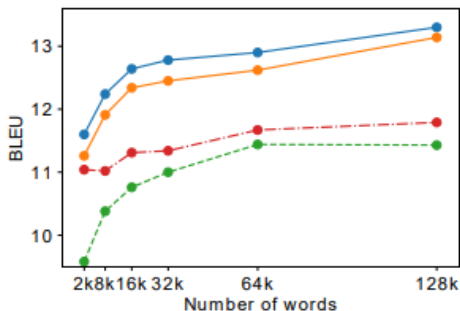
Corpus	Words		Sentences	W/S	
	EN	DE		EN	DE
Acquis (Law)	9.2M	8M	0.7M	12.93	11.30
EMEA (Medical)	7.5M	6.3M	1.1M	6.81	5.75
IT	1.7M	1M	0.3M	9.08	5.32
Tanzil (Koran)	5.6M	5.3MS	0.5M	10.66	10.08
Subtitles	92.7M	87.6M	22.5M	4.11	3.89
EUbookshop (EUB)	115.4M	100M	9.3M	12.37	10.72
Europarl	27.3M	25.7M	1.9M	13.99	13.18
GlobalVoices (GV)	0.6M	0.6M	0.05M	10.67	10.88

Table T.2: Statistics of each corpora.

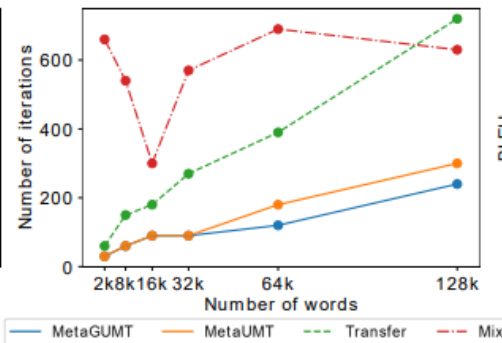
Experiments

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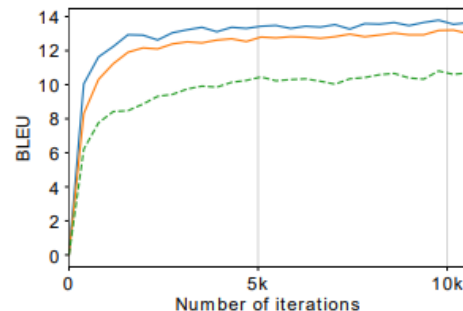
- Experimental Settings
 - Transformer from XLM (Conneau and Lample, 2019)
 - 6 layers, 1,024 units, and 8 heads.
- Experimental Results



(A)



(B)



(C)

Experiments

03

- Performances and Adaptation Speed in Finetuning Stage

Parameter	Initial θ												Finetuned θ			
D	$\mathcal{D}_{out}$												$\mathcal{D}_{in}$		Unseen	
	Meidcal		Law		Koran		EUB		IT		GV		Subtitles		Europarl	
	De-En	En-De	De-En	En-De	De-En	En-De	De-En	En-De	De-En	En-De	De-En	En-De	De-En	En-De	De-En	En-De
Transfer	30.98	26.96	34.8	30.28	13.72	11.59	12.32	10.01	20.98	17.74	17.4	14.25	10.92	9.18	22.31	16.58
Mixed finetuned	-	-	-	-	-	-	-	-	-	-	-	-	11.77	9.96	22.84	16.92
MetaUMT	33.0	23.39	27.74	15.4	4.89	0.79	6.78	2.59	9.45	4.68	2.77	1.06	12.95	10.58	23.91	18.7
MetaGUMT	37.37	31.63	42.73	37.3	18.2	13.84	13.72	11.8	24.0	19.24	21.24	17.38	13.45	10.89	24.44	19.31

- Analysis of MetaGUMT losses

Cross-domain	Aggregated meta-train	De-En	En-De	Average	Δ
$\times$	$\times$	27.09	24.6	25.85	
$\checkmark$	$\times$	27.37	24.76	26.06	+0.21
$\times$	$\checkmark$	27.54	24.90	26.22	+0.37
$\checkmark$	$\checkmark$	27.85	25.06	26.46	+0.61

Experiments

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- Performance of Unbalanced Monolingual Data in Finetuning Stage

# tokens		Mixed		MetaUMT		MetaGUMT	
En	De	En-De	De-En	En-De	De-En	En-De	De-En
5k	10k	26.04	31.90	28.80	32.65	29.43	34.28
8k	16k	26.09	32.01	27.84	32.93	29.62	34.39
16k	32k	26.44	32.37	27.92	32.96	30.10	34.44
32k	64k	27.39	32.84	28.67	33.52	29.83	34.77

- Impact of the Number of Source Domains

# $\mathcal{D}_{out}$	MetaGUMT		MetaUMT		Transfer		Mixed	
	En-De	De-En	En-De	De-En	En-De	De-En	En-De	De-En
4	5.97	7.47	5.87	7.24	5.75	7.17	5.87	7.22
5	7.58	9.49	7.33	9.01	7.17	8.08	7.20	8.68
6	10.89	13.45	10.58	12.95	9.18	10.92	9.96	11.77

Implementation Details

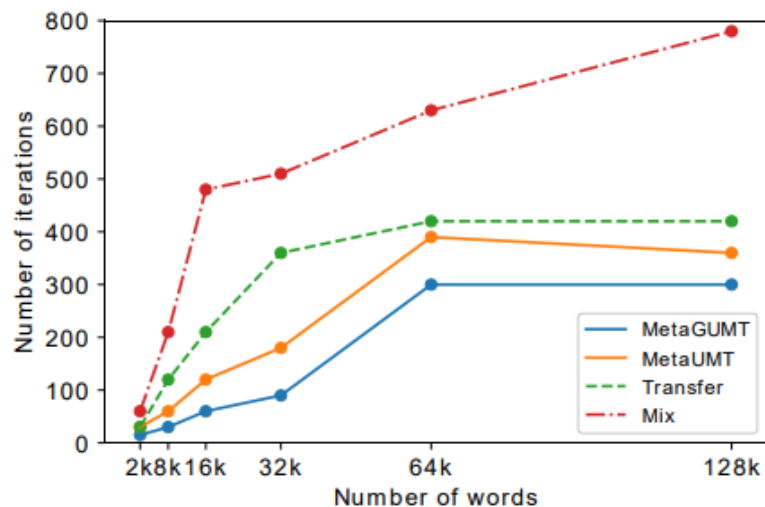
03

- Moses (Koehn et al., 2007) to tokenize the sentences
- use byte-pair encoding (BPE)
- sub-word vocabulary using fastBPE7 with 60,000 BPE codes
- using PyTorch library 8 | four nvidia V100 gpus
- on the BLEU script 9 | on the best validation epoch + 10 more epochs
- The learning rate = 10^{-4} | optimized within the range of 10^{-2} to 10^{-5}
- number of tokens per batch = 1,120 | dropout rate = 0.1

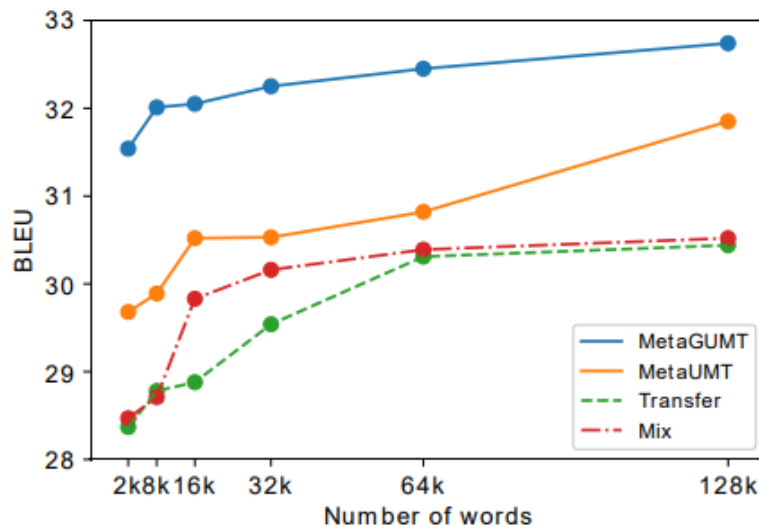
Implementation Details

03

- Additional Results on Different Domain Combinations



Number of iterations until the convergence



A performance comparison

- proposes a novel meta-learning approach for low-resource UNMT, called MetaUMT
- MetaGUMT : enhances cross-domain generalization and maintains high-resource domain knowledge
- can be extended to semi-supervised machine translation

NLP Team Review Opinion

04

- 적은 도메인 자원을 해결하기 위한 새로운 제안으로 주목할 만 함.
- 기대했던 semi-supervised machine translation의 확장된 후속 제안이 아직은 활발하지 않은 것이 아쉬웠다.
- 기존의 전이 학습과 제안된 Meta-learning 방법 간의 비교 실험이 없었던 것이 아쉬웠다.

Q & A

Thank you!